

SOFTWARE ENGINEERING

Lab3:

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Component Modelling & Architectural Pattern Selection:

Project: Group Travel Expense Simplification Engine

Architecture Selected: Layered Architecture

Architecture Selection:

Layered Architecture for the Group Travel Expense Simplification Engine.

Separation of Responsibilities

The system has clearly separated responsibilities such as user interaction, trip and expense management, currency conversion, balance calculation, settlement processing, and data storage. Layered Architecture separates these responsibilities into Presentation, Application, Business, and Data layers, making the system easier to understand, maintain, and modify.

Suitable for the Expense and Settlement Workflow

The Group Travel Expense Simplification Engine follows a clear processing flow in which expenses are recorded, foreign currencies are converted, balances are calculated, and the final settlement plan is generated. Layered Architecture supports this sequential flow by keeping application operations separate from business logic and database operations.

Security Advantage

Layered Architecture provides separation between the user interface, application logic, business logic, and data layer. Access control and authentication can be enforced before users reach expense, balance, and settlement operations, helping restrict trip and financial data to authorized trip members.

Performance Benefit

The architecture separates the computationally intensive balance calculation and settlement processing from the presentation layer. This allows the settlement logic to execute directly within the business layer without placing unnecessary processing responsibilities on the user

interface, supporting efficient settlement-plan generation for the specified group and expense sizes.